3.4 Tensor Products

3.4.1 Commutative Tensor Product

We consider the "Tensor Product" of modules over a single commutative ring as an important special case, and as a guide to the general bimodule case.

In order to streamline the (typically tedious) verification of standard properties we make heavy use of the notion of **representable bifunctor** (2.6.65) however all the results may be proved in an essentially identical "low-tech" way.

First we define the so-called "Internal Hom Functor"

Definition 3.4.1 (Internal Hom)
Let A be a commutative ring and M, N be A-modules. The set of A-linear homomorphisms $M \to N$ is itself an A-module. Therefore we have an enriched hom functor

$$\mathrm{Hom} : A\text{-}\mathbf{Mod}^{op} \times A\text{-}\mathbf{Mod} \to A\text{-}\mathbf{Mod}$$

where

$$(a \cdot \psi)(m) \coloneqq a \cdot \psi(m) = \psi(a \cdot m)$$

and in particular we have a bijection

$$\mathrm{Mor}(M, N) \xrightarrow{\sim} \mathrm{Nat}(\mathrm{Hom}(N, -), \mathrm{Hom}(M, -))$$

Definition 3.4.2 (Bilinear Map)
Let M, N, Z be A-modules. Then a map

$$\psi : M \times N \to Z$$

*is **bilinear** if it satisfies*

- ***additive*** $\psi(m + m', n) = \psi(m, n) + \psi(m', n)$ *and* $\psi(m, n + n') = \psi(m, n) + \psi(m, n')$
- ***A-linear*** $\psi(am, n) = \psi(m, an) = a\psi(m, n)$ *for all* $a \in A$

This yields a bifunctor which we denote by

$$\mathrm{Bilin}(-, -; -) : (A\text{-}\mathbf{Mod} \times A\text{-}\mathbf{Mod})^{op} \times A\text{-}\mathbf{Mod} \to A\text{-}\mathbf{Mod}$$

Remark 3.4.3 (A-linearity)
The A-module structure on the image of Hom *and* Bilin *is only well-defined because A is commutative. For given $\psi : M \to N$ an A-linear map then for $(a \cdot \psi)$ to be A-linear we may verify this requires*

$$a'a\psi(m) = (a\psi)(a'm) = aa'\psi(m)$$

which in general does not hold unless A is commutative.

Proposition 3.4.4 (Existence of Tensor Product)
The bifunctor $\mathrm{Bilin}(-, -; -)$ *is representable by a functor*

$$\otimes_A : A\text{-}\mathbf{Mod} \times A\text{-}\mathbf{Mod} \to A\text{-}\mathbf{Mod}$$

for which there exists an isomorphism natural in M, N, Z

$$\Phi : \mathrm{Hom}(M \otimes N, Z) \cong \mathrm{Bilin}(M, N; Z)$$

For every pair M, N there is a distinguished bilinear map

$$i : M \times N \to M \otimes N$$

*through which every bilinear map $M \times N \to Z$ factors uniquely. We denote the **elementary tensor** by*

$$m \otimes n \coloneqq i(m, n)$$

Further for morphisms $f : M \to M'$ and $g : N \to N'$ there is a unique morphism

$$(f \otimes g) : M \otimes N \to M' \otimes N'$$

such that

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$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n)$$

Proof. In light of (2.6.66) it is sufficient to show that $\mathrm{Bilin}(M, N; -)$ is representable for fixed M, N and the rest of the properties follow, observing that $i = \Phi(1_{M \otimes N})$.

For let F be the free abelian group on $M \times N$ and let $S \subset F$ be the subgroup generated by elements of the form

- $(m + m', n) - (m, n) - (m', n)$
- $(m, n' + n) - (m, n') - (m, n)$
- $(am, n) - a(m, n)$
- $(m, n) - (m, an)$

Observe there is a canonical map $M \times N \to F$. Suppose that Z is an abelian group and $\psi : M \times N \to Z$ is a bilinear map. Then we may extend by linearity to a group homomorphism $\psi : F \to Z$. By construction $S \subseteq \ker(\psi)$ whence there is a unique group homomorphism $\hat{\psi} : F/S \to Z$ such that $\hat{\psi} \circ i = \psi$ where $i : M \times N \to F/S$ is the canonical inclusion. This shows that $(F/S, i)$ is a universal element for $\mathrm{Bilin}(-, -; -)$ as we required and we denote this by $M \otimes N$. The representation exists as a functor of sets by (2.6.62), and Φ is clearly an isomorphism of A-modules because

$$\Phi(f + g) = (f + g) \circ i = f \circ i + g \circ i = \Phi(f) + \Phi(g)$$

and

$$\Phi(\lambda f) = (\lambda f) \circ i = \lambda(f \circ i) = \lambda \Phi(f)$$

■

Lemma 3.4.5 (Currying Lemma)
Let M, N, Z be A-modules then there are natural isomorphisms

$$\begin{array}{ccccccc} \mathrm{Bilin}(M, N, Z) & \cong & \mathrm{Hom}(M, \mathrm{Hom}(N, Z)) & \cong & \mathrm{Hom}(N, \mathrm{Hom}(M, Z)) & \cong & \mathrm{Bilin}(N, M, Z) \\ \psi & & m \to (n \to \psi(m, n)) & & n \to (m \to \psi(m, n)) & & (n, m) \to \psi(m, n) \end{array}$$

This gives the following important result (which may be seen as an adjoint relationship)

Proposition 3.4.6 (Tensor-Hom Adjunction)
Let M, N, Z be A-modules then we have the following natural isomorphism of functors

$$\begin{array}{rl} \mathrm{Hom}(M \otimes N, Z) \cong & \mathrm{Hom}(M, \mathrm{Hom}(N, Z)) \\ & \theta \to (m \to (n \to \theta(m \otimes n))) \end{array}$$

We may also use the Currying Lemma (3.4.5) to demonstrate symmetry of the tensor product

Proposition 3.4.7 (Symmetry of Tensor Product)
There is a natural isomorphism

$$\begin{array}{rl} M \otimes N \cong & N \otimes M \\ m \otimes n \to & n \otimes m \end{array}$$

Proof. We observe that there is a natural isomorphism of functors which the tensor products represent

$$\mathrm{Bilin}(M, N; Z) \cong \mathrm{Bilin}(N, M; Z)$$

so the result follows from (2.6.68). ■

Proposition 3.4.8 (Associativity of Tensor Product)
There is a natural isomorphism of A-modules

$$\begin{array}{rl} (M \otimes N) \otimes P \cong & M \otimes (N \otimes P) \\ (m \otimes n) \otimes p \to & m \otimes (n \otimes p) \end{array}$$

Proof. We may make repeated use of the Tensor-Hom adjunction to exhibit natural isomorphisms

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$$\begin{aligned}\mathrm{Hom}((M\otimes N)\otimes P, Z) &\cong \mathrm{Hom}(M\otimes N, \mathrm{Hom}(P,Z))\\ &\cong \mathrm{Hom}(M, \mathrm{Hom}(N, \mathrm{Hom}(P,Z)))\\ &\cong \mathrm{Hom}(M, \mathrm{Hom}(N\otimes P, Z))\\ &\cong \mathrm{Hom}(M\otimes(N\otimes P), Z)\end{aligned}$$

The required isomorphism is then a consequence of (2.6.68). For explicit form set $Z = M\otimes(N\otimes P)$ and consider the identity map 1_Z. We see tracing back under these natural isomorphisms this corresponds to the given map. ■

3.4.2 Bimodule Tensor Product

To develop the tensor product in general it is convenient to work with bimodules. We make heavy use of the notion if bimodule hom-sets (3.3.78).

Definition 3.4.9 (Bilinear maps)
Consider the bimodules ${}_AM_B$, ${}_BN_C$. We say that a map

$$\psi : {}_AM_B \times {}_BN_C \to {}_AZ_C$$

*is (A,C)-**bilinear** if it satisfies all of the following conditions*

- ***additive*** $\psi(m+m', n+n') = \psi(m,n)+\psi(m',n)+\psi(m,n')+\psi(n,n')$
- ***balanced*** $\psi(mb,n) = \psi(m,bn)$
- ***A-linear*** $\psi(am,n) = a\psi(m,n)$
- ***C-linear*** $\psi(m,nc) = \psi(m,n)c$

Extending the notation from earlier we denote the set of additive, balanced and (A,C)-bilinear maps by $\mathrm{Bilin}(M,N;Z)$. *This determines a functor*

$$\mathrm{Bilin} : ({}_A\mathbf{Mod}_B \times {}_B\mathbf{Mod}_C)^{op} \times {}_A\mathbf{Mod}_C \to \mathbf{AbGrp}$$

where addition is defined pointwise. We may also consider the functor of only additive and balanced maps

$$\mathrm{Balan} : (\mathbf{Mod}_B \times {}_B\mathbf{Mod})^{op} \times \mathbf{AbGrp} \to \mathbf{AbGrp}$$

and the functor

$$\mathrm{RBilin} : ({}_B\mathbf{Mod}_C \times {}_C\mathbf{Mod}_D)^{op} \times {}_A\mathbf{Mod}_D \to {}_A\mathbf{Mod}_B$$

where we drop the B-linear requirement but also introduce an extra (A,B)-module structure on the maps, as follows

$$(a\psi b)(m,n) \coloneqq a\psi(bm,n)$$

Proposition 3.4.10 (Existence of Bimodule Tensor Product)
Let A, B, C be arbitrary rings

- *The bifunctor* $\mathrm{Balan}(M,N;-)$ *is represented by the **balanced tensor product***

  $$\otimes_B : \mathbf{Mod}_B \times {}_B\mathbf{Mod} \to \mathbf{AbGrp}$$

  with natural isomorphism for Z an Abelian group

  $$\begin{aligned}\Phi : \mathrm{Hom}(M_B \otimes_B {}_BN, Z) &\cong \mathrm{Balan}(M_B, {}_BN; Z)\\ \theta &\to \theta\circ i\end{aligned}$$

  where i is a universal balanced map

  $$i : M\times N \to M_B\otimes_B {}_BN$$

  *The tensor product is generated by **elementary tensors** of the form $m\otimes n \coloneqq i(m,n)$.*

- *The bifunctor* $\mathrm{Bilin}(M,N;-)$ *is represented by the balanced tensor product $M\otimes_B N$ with an additional (A,C)-bimodule structure given by*

  $$a\cdot(m\otimes n)\cdot c \coloneqq (a\cdot m)\otimes(n\cdot c)$$

  This determines a bifunctor

  $$\otimes_B : {}_A\mathbf{Mod}_B \times {}_B\mathbf{Mod}_C \to {}_A\mathbf{Mod}_C$$

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and a natural isomorphism of Abelian groups

$$\begin{aligned}\mathrm{Hom}({}_AM_B \otimes_B {}_BN_C, Z) &\cong \mathrm{Bilin}({}_AM_B, {}_BN_C; {}_AZ_C)\\ \theta &\to \theta \circ i\end{aligned}$$

In either case, if $f : M \to M'$ and $g : N \to N'$ are morphisms then there is a unique morphism

$$(f \otimes g) : (M \otimes N) \to (M' \otimes N')$$

such that

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n)$$

Furthermore for maps $f' : M' \to M''$ and $g' : N' \to N''$ the following property is satisfised

$$(f' \otimes g') \circ (f \otimes g) = (f' \circ f) \otimes (g' \circ g)$$

Proof. In light of (2.6.65) we only need to show that the functors are representable for fixed M, N, as the functoriality will follow immediately by abstract nonsense.

We first show that $\mathrm{Balan}(-,-;-)$ is representable. For let F be the free abelian group on $M \times N$ and let $S \subset F$ be the subgroup generated by elements of the form

- $(m + m', n) - (m, n) - (m', n)$
- $(m, n' + n) - (m, n') - (m, n)$
- $(mb, n) - (m, bn)$

Observe there is a canonical map $M \times N \to F$. Suppose that Z is an abelian group and $\psi : M \times N \to Z$ is a balanced additive map. Then we may extend by linearity to a group homomorphism $\psi : F \to Z$. By construction $S \subseteq \ker(\psi)$ whence there is a unique group homomorphism $\hat{\psi} : F/S \to Z$ such that $\hat{\psi} \circ i = \psi$ where $i : M \times N \to F/S$ is the canonical inclusion. This shows that $(F/S, i)$ is a universal element for $\mathrm{Balan}(-,-;-)$ and we denote this by $M_B \otimes_B {}_BN$. Then (2.6.62) exhibits Φ as a natural isomorphism of sets, which may readily observed to be additive.

In order to define the (A, C)-bimodule structure in the second case let $\psi_a : M \to M$ denote multiplication by a and $\psi_c : N \to N$ multiplication by c. We note these are both B-module homomorphisms by definition so we may define the action by functoriality of the balanced tensor product case

$$\begin{aligned}a \cdot v &:= (\psi_a \otimes 1_N)(v) \quad \forall v \in M \otimes N\\ v \cdot c &:= (1_M \otimes \psi_c)(v) \quad \forall v \in M \otimes N\end{aligned}$$

These are B-linear by construction and we may demonstrate (A, C)-bimodule actions are associative and commute by using the functoriality of the tensor product

$$\begin{aligned}(\psi_{a'} \otimes 1_N) \circ (\psi_a \otimes 1_N) = (\psi_{a'} \circ \psi_a) \otimes 1_N &= (\psi_{a'a}) \otimes 1_N\\ (1_M \otimes \psi_c) \circ (1_M \otimes \psi_{c'}) = 1_M \otimes (\psi_c \circ \psi_{c'}) &= 1_M \otimes \psi_{cc'}\\ (\psi_a \otimes 1_N) \circ (1_M \otimes \psi_c) = \psi_a \otimes \psi_c &= (1_M \otimes \psi_c) \otimes (\psi_a \otimes 1_N)\end{aligned}$$

Further by construction the canonical map i is (A, C)-bilinear. Consider any (A, C)-bilinear map $\psi : M \times N \to Z$ then by the previous part there is a unique group homomorphism $\hat{\psi} : M \otimes_B N \to Z$ such that

$$\hat{\psi}(m \otimes n) := \psi(m, n)$$

which we see by construction (and linearity) is (A, C)-bilinear. It is clearly the unique such map so that we may conclude $\mathrm{Bilin}(-,-;-)$ has universal element $({}_AM_B \otimes_B {}_BN_C, i)$. The (2.6.62) exhibits Φ as a natural isomorphism of sets which may be readily exhibited to be additive. ∎

Lemma 3.4.11 (Currying Lemma)
Let M, N, P, Y, Z be bimodules. Then there is an isomorphism of abelian groups and (A, B)-bimodules respectively

$$\begin{aligned}\mathrm{Bilin}({}_AM_B, {}_BN_C; {}_AY_C) &\cong \mathrm{Hom}({}_AM_B, \mathrm{RHom}({}_BN_C, {}_AY_C))\\ &\cong \mathrm{Hom}({}_BN_C, \mathrm{LHom}({}_AM_B, {}_AY_C))\\ \mathrm{RBilin}({}_BN_C, {}_CP_D; {}_AZ_D) &\cong \mathrm{RHom}({}_BN_C, \mathrm{RHom}({}_CP_D, {}_AZ_D))\end{aligned}$$

contravariant in M, N, P and covariant in Y and Z.

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Proof. The maps on abelian groups are clear, namely given a bi-additive map ψ we have a well-defined homomorphism of abelian groups

$$m \to (n \to \psi(m,n))$$

The verification that the maps are well-defined and bijective is tedious but mechanical. ∎

Proposition 3.4.12 (Tensor-Hom Adjunction)
Let M, N, P, Y, Z be bimodules then

$$\begin{aligned} \mathrm{Hom}({}_AM_B \otimes {}_BN_C, {}_AY_C) &\cong \mathrm{Hom}({}_AM_B, \mathrm{RHom}({}_BN_C, {}_AY_C)) \\ &\cong \mathrm{Hom}({}_BN_C, \mathrm{LHom}({}_AM_B, {}_AY_C)) \\ \mathrm{RHom}({}_BN_C \otimes {}_CP_D, {}_AZ_D) &\cong \mathrm{RHom}({}_BN_C, \mathrm{RHom}({}_CP_D, {}_AZ_D)) \end{aligned}$$

where the first is an isomorphism of Abelian Groups and the second of (A,B)-bimodules. Further the isomorphisms are contravariant in M, N, P and covariant in Y and Z.

Proof. This first two isomorphism follow directly from the Currying Lemma and universal property for bimodule tensor product.

The final isomorphism follows from the Currying Lemma if we show that there is an isomorphism

$$\begin{array}{ccc} \mathrm{RHom}({}_BN_C \otimes {}_CP_D, {}_AZ_D) & \xrightarrow{\sim} & \mathrm{RBilin}({}_BN_C, {}_CP_D; {}_AZ_D) \\ \downarrow & & \downarrow \\ \mathrm{Hom}(N_C \otimes {}_CP, Z) & \xrightarrow{\sim} & \mathrm{Balan}(N_C, {}_CP; Z) \end{array}$$

where the bottom is simply the universal property of the balanced tensor product and the top we require to also be (A,B)-bilinear. Recall the underlying sets are the same and the horizontal maps are given by $\theta \to \theta \circ i$ in both cases. It's clear that the map is well-defined and injective. Further it's surjective because if θ is (B,D)-bilinear on elementary tensors it is (B,D)-bilinear everywhere. Finally we need to show that the isomorphism is (A,B)-bilinear as follows

$$(a(\theta \circ i)b)(n,p) = a(\theta \circ i)(bn,p) = a\theta((bn) \otimes p) = a\theta(b(n \otimes p)) = (a\theta b)(n \otimes p)$$

∎

Proposition 3.4.13 (Associativity of Tensor Product)
Let ${}_AM_B$, ${}_BN_C$, ${}_CP_D$ be bimodules then there is a natural isomorphism of (A,D)-bimodules

$$\begin{aligned} (M \otimes_B N) \otimes_C P &\cong M \otimes_B (N \otimes_C P) \\ (m \otimes n) \otimes p &\to m \otimes (n \otimes p) \end{aligned}$$

Proof. This follows much as in the commutative case. Let Z be an (A,D)-bimodule. Then by the Tensor-Hom adjunction there are natural isomorphisms of Abelian groups

$$\begin{aligned} \mathrm{Hom}(({}_AM_B \otimes {}_BN_C) \otimes {}_CP_D, {}_AZ_D) &\cong \mathrm{Hom}({}_AM_B \otimes {}_BN_C, \mathrm{RHom}({}_CP_D, {}_AZ_D)) \\ &\cong \mathrm{Hom}({}_AM_B, \mathrm{RHom}({}_BN_C, \mathrm{RHom}({}_CP_D, {}_AZ_D))) \\ &\cong \mathrm{Hom}({}_AM_B, \mathrm{RHom}({}_BN_C \otimes_C {}_CP_D, {}_AZ_D)) \\ &\cong \mathrm{Hom}({}_AM_B \otimes_B ({}_BN_C \otimes_C {}_CP_D), {}_AZ_D) \end{aligned}$$

The required isomorphism is then a consequence of (2.6.68). For explicit form set $Z = M \otimes (N \otimes P)$ and consider the identity map 1_Z. We see tracing back under these natural isomorphisms this corresponds to the given map. ∎

3.4.3 Extensions of Scalars

We observe that for a ring homomorphism $\varphi : A \to B$ we have on B a natural (A,B)-bimodule and (B,A)-bimodule structure. Using the tensor product then we may use this to extend the coefficients from A to B. We first prove some elementary lemmas

Lemma 3.4.14 (Unit Hom)
Let $\varphi : B \to C$ be a ring homomorphism and Z a (A,C)-bimodule. Then there is a natural isomorphism of (A,B)-bimodules

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$$\begin{aligned}\mathrm{RHom}({}_BC_C, {}_AZ_C) &\cong {}_AZ_B\\ \theta &\to \theta(1)\\ z(-) &\leftarrow z\end{aligned}$$

and similarly let $\varphi : A \to B$ be a ring homomorphism and Y a (B,C)-bimodule then there is a natural isomorphism of (A,C)-bimodules

$$\begin{aligned}\mathrm{LHom}({}_BB_A, {}_BY_C) &\cong {}_AY_C\\ \theta &\to \theta(1)\end{aligned}$$

Proof. Observe

$$(a\theta b)(1) = a\theta(\varphi(b)) = a\theta(1)\varphi(b)$$

so the left to right map is an (A,B)-bimodule homomorphism. Further $\theta(c) = \theta(1)c$ so the maps are mutual inverses.■

Lemma 3.4.15 (Unit Hom (Commutative Ring Case))
Let $\varphi : A \to B$ be a homomorphism of commutative rings and Z a B-module. Then there is a natural isomorphism of B-modules

$$\begin{aligned}\mathrm{Hom}(B, Z) &\cong Z\\ \theta &\to \theta(1)\end{aligned}$$

Proposition 3.4.16 (Extension of Scalars)
Let M be an (A,C)-bimodule and $\varphi : A \to B$ a ring homomorphism. We may define a (B,C)-module

$$M_{(B)} \coloneqq {}_BB_A \otimes_A M$$

For Z a (B,C)-bimodule there is a natural isomorphism of abelian groups

$$\begin{aligned}\mathrm{Hom}\Big(M_{(B)}, Z\Big) &\xrightarrow{\sim} \mathrm{Hom}(M, {}_AZ_C)\\ \psi &\to m \to \psi(1 \otimes m)\\ b \otimes m \to b \cdot \phi(m) &\leftarrow \phi\end{aligned}$$

In otherwords we have an adjunction (2.6.69)

$${}_A\mathbf{Mod}_C \longleftrightarrow {}_B\mathbf{Mod}_C$$

Setting $C = \mathbb{Z}$ yields an adjunction for left modules

$${}_A\mathbf{Mod} \longleftrightarrow {}_B\mathbf{Mod}$$

In particular for M a left A-module we have an isomorphism of abelian groups

$$\mathrm{Hom}\Big(M_{(B)}, B\Big) \xrightarrow{\sim} \mathrm{Hom}_A(M, B)$$

which is B-linear when B is commutative.

Proof. By the tensor-hom adjunction (3.4.12) and (3.4.14) there is a natural isomorphism

$$\mathrm{Hom}(B \otimes_A M, Z) \cong \mathrm{Hom}(M, \mathrm{LHom}({}_BB_A, {}_BZ_C)) \cong \mathrm{Hom}(M, {}_AZ_C)$$

■

Proposition 3.4.17
Let $\varphi : A \to B$ be a homomormorphism of commutative rings and $\varphi : M \to N$ a homomorphism of A-modules. Then there is a unique B-module homomorphism $\varphi_{(B)} : M_{(B)} \to N_{(B)}$ such that

$$\varphi_{(B)}(1 \otimes m) = 1 \otimes \varphi(m)$$

Proof. The homomorphism $\varphi_{(B)} = 1_B \otimes \psi$ exists by functoriality of the tensor product (3.4.4). It clearly satisfies the given property, and as the tensor product is generated by elementary tensors it is evidentally unique. ■

Proposition 3.4.18 (Tensor Unit)
Let M be an (A,C)-bimodule then there is a natural isomorphism of (A,C)-bimodules

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$$\begin{aligned} A \otimes_A M &\cong M \\ a \otimes m &\to a \cdot m \\ 1 \otimes m &\leftarrow m \end{aligned}$$

Similarly there is a natural isomorphism of (A, C)-bimodules

$$M \otimes_C C \cong M$$

Proof. By (3.4.16) with $\phi = 1_A$ there is a natural isomorphism

$$\mathrm{Hom}(A \otimes_A M, -) \cong \mathrm{Hom}(M, -)$$

and so natural isomorphism follows from (2.6.68).

We may deduce that there is a natural isomorphism of $(C^{\mathsf{op}}, A^{\mathsf{op}})$-bimodules

$$C^{\mathsf{op}} \otimes_{C^{\mathsf{op}}} M^{\mathsf{op}} \cong M^{\mathsf{op}}$$

which amounts to a natural isomorphism of (A, C)-bimodules

$$M \otimes_C C \cong M$$

■

Proposition 3.4.19 (Transitivity of Extension of Scalars)
Let M be a (A, D)-bimodule and $\varphi : A \to B$, $\psi : B \to C$ homomorphisms of commutative rings. Then there is an isomorphism of (C, D)-bimodules

$$\begin{aligned} C \otimes_B (B \otimes_A M) &\xrightarrow{\sim} {}_C C_A \otimes_A M \\ c \otimes (b \otimes m) &\to (c\psi(b)) \otimes m \end{aligned}$$

Proof. By associativity (3.4.13) and (3.4.18) there is a natural isomorphism

$$C \otimes_B (B \otimes_A M) \cong (C \otimes_B B) \otimes_A M \cong C \otimes_A M$$

■

Proposition 3.4.20 (Transitivity of Extension of Scalars (Commutative Case))
Let M be an A-module and $\phi : A \to B$ and $\psi : B \to C$ homomorphisms of commutative rings. Then there is an isomorphism of C-modules

$$C \otimes_B (B \otimes_A M) \cong {}_C C_A \otimes_A M$$

Proof. Regarding M as an (A, A)-bimodule we have an isomorphism of (C, A)-bimodules by (3.4.19) which is a-fortiori a C-module isomorphism. ■

3.4.4 Tensor Product Commutes with Direct Sum

Proposition 3.4.21 (Tensor Product Commutes with Sum)
Let $(M_i)_{i \in I}$ be a family of (A, B)-bimodules and $(N_j)_{j \in J}$ a family of (B, C)-bimodules. Then there is an isomorphism of (A, C)-bimodules

$$\left(\bigoplus_{i \in I} M_i \right) \otimes_B \left(\bigoplus_{j \in J} N_j \right) \cong \bigoplus_{(i,j) \in I \times J} (M_i \otimes_B N_j)$$

Corollary 3.4.22
Suppose M is an (A, B)-bimodule and N is a (B, C)-bimodule and $M' \subseteq M$ and $N' \subseteq N$ are direct factors. Then the canonical map is injective

$$M' \otimes_B N' \hookrightarrow M \otimes_B N$$

Corollary 3.4.23
*Let N be **free** left A-module with basis $(n_i)_{i \in I}$ and M a (B, A)-bimodule. Then there is an isomorphism of left B-modules*

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$$M \otimes_A N \cong \bigoplus_{i\in I} M$$
$$m \otimes \sum_i a_i n_i \to (m \cdot a_i)_{i\in I}$$
$$\sum_i (m_i \otimes n_i) \leftarrow (m_i)_{i\in I}$$

When A is commutative then this is a (B,A)-bimodule isomorphism. Further when M is an A-module then this is an isomorphism of A-modules.

Proof. By (3.4.21) and (3.4.18)

$$M \otimes_A N \cong M \otimes_A \left(\bigoplus_{i\in I} A\right) \cong \bigoplus_{i\in I}(M \otimes_A A) \cong \bigoplus_{i\in I} M$$
$$m \otimes \sum_i a_i n_i \to \quad m \otimes (a_i)_{i\in I} \to \quad (m \otimes a_i)_{i\in I} \to (m \cdot a_i)_{i\in I}$$

■

Corollary 3.4.24
If M, N are free A-modules with bases $\{m_i\}_{i\in I}$ and $\{n_j\}_{j\in J}$ then $M \otimes_A N$ is a free A-module with basis $\{m_i \otimes n_j\}_{(i,j)\in I\times J}$.

Corollary 3.4.25 (Free modules are flat)
Let N be a free left A-module and $i : M' \to M$ an injective map of (B,A)-bimodules. Then the corresponding map

$$i \otimes 1_N : M' \otimes_A N \to M \otimes_A N$$

is injective.

Corollary 3.4.26 (Extension of Scalars (Free Module))
*Let M be **free** left A-module with basis $\{m_i\}_{i\in I}$ and $\varphi : A \to B$ a ring homomorphism. Then there is a canonical isomorphism of left B-modules*

$$M_{(B)} \cong \bigoplus_{i\in I} B$$
$$b \otimes \sum_i a_i m_i \to (b\varphi(a_i))_{i\in I}$$

In particular $\{1 \otimes m_i\}_{i\in I}$ is a basis for $M_{(B)}$. Further when Z is a left B-module then there is an isomorpihsm of abelian groups

$$\begin{array}{ccc} \mathrm{Hom}_A(M,Z) & \cong & \mathrm{Hom}_B\big(M_{(B)},Z\big) \cong \prod_{i\in I} Z \\ \theta & & \to \quad (\theta(m_i))_{i\in I} \end{array}$$

which is B-linear when B is commutative.

When M is a finite free A-module of rank n (with basis $\{v_i\}$) then $M_{(B)}$ is a finite free B-module of rank n (with basis $\{1 \otimes v_i\}$). When B is commutative, $M_{(B)}^\vee$ is also a finite free B-module of rank n.

Proof. The first isomorphism follows directly from (3.4.23). Then by (3.3.82) there is an isomorphism

$$\mathrm{Hom}\big(M_{(B)},Z\big) \cong \mathrm{Hom}\left(\bigoplus_{i\in I} B, Z\right) \cong \prod_{i\in I} \mathrm{Hom}(B,Z) \cong \prod_{i\in I} Z$$

■

We may generalize (3.3.109)

Proposition 3.4.27 (Extension of Scalars (Hom-Set))
Let $\varphi : A \to B$ be a homomorphism of commutative rings, M a finite-free A-module and N a finite-free B-module. Then

$$\mathrm{Hom}_A(M,N) \cong \mathrm{Hom}_B\big(M_{(B)},N\big)$$

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is a finite-free B-module. More precisely suppose M has basis $\{v_1,...,v_n\}$ and N has basis $\{w_1,...,w_m\}$ then $\mathrm{Hom}_A(M,N)$ *has basis*

$$\left\{w_j v_i^\vee\right\}_{i,j}$$

Proof. By (3.4.26) there are isomorphisms of B-modules

$$\begin{aligned}\mathrm{Hom}_A(M,N) \cong \mathrm{Hom}_B\left(M_{(B)},N\right) \cong \bigoplus_{i=1}^{n} N \cong \bigoplus_{i=1}^{n}\bigoplus_{j=1}^{m} B\\ \theta \to \qquad 1\otimes\theta \to (\theta(v_i))_i \to \left(w_j^\vee(\theta(v_i))\right)_{i,j}\end{aligned}$$

Under this isomorphism the standard basis of the right hand side corresponds to the set $\left\{w_j v_i^\vee\right\}_{i,j}$ whence these constitute a basis. ∎

3.4.5 Tensor Product Exact Sequences

Proposition 3.4.28 (Tensor Product is Right Exact)
Consider an exact sequence of A-modules (i.e. such that β is surjective and $\ker(\beta)=\mathrm{Im}(\alpha)$*)*

$$M \xrightarrow{\alpha} N \xrightarrow{\beta} P \to 0$$

Then the corresponding sequence of A-modules

$$M\otimes_A Z \xrightarrow{\alpha\otimes 1_Z} N\otimes_A Z \xrightarrow{\beta\otimes 1_Z} P\otimes_A Z \to 0$$

is exact. Similarly the sequence

$$Z\otimes_A M \xrightarrow{1_Z\otimes\alpha} Z\otimes_A N \xrightarrow{1_Z\otimes\beta} Z\otimes_A P \to 0$$

is exact.

Proof. Trivially the submodule $\mathrm{Im}(\beta\otimes 1_Z)$ contains elementary tensors and therefore equals $P\otimes_A Z$. In otherwords $\beta\otimes 1_Z$ is surjective. Furthermore by functoriality $(\beta\otimes 1_Z)\circ(\alpha\otimes 1_Z)=(\beta\circ\alpha)\otimes 1_Z$ which is zero on elementary tensors and therefore zero everywhere. In particular $\mathrm{Im}(\alpha\otimes 1_Z)\subseteq\ker(\beta\otimes 1_Z)$. Consider a commutative diagram

$$\begin{array}{ccc} P\times Z & \overset{\psi}{\dashrightarrow} & (N\otimes_A Z)/\mathrm{Im}(\alpha\otimes 1_Z) \\ & \searrow & \downarrow{\scriptstyle \overline{\beta\otimes 1_Z}} \\ & & P\otimes_A Z \end{array}$$

where ψ is yet to be defined. For a given $p\in P$ and $z\in Z$ consider the subset

$$\mathcal{I}_{p,z} \coloneqq \{n\otimes z \mid n\in N, \beta(n)=p\}\subset N\otimes_A Z$$

Observe that any ψ making the diagram commute satisfies $\psi(p,z)\in\overline{\mathcal{I}_{p,z}}$, and we aim to show this set is a singleton.

As β is surjective $\mathcal{I}_{p,z}$ is non-empty. First we show that $x,y\in\mathcal{I}_{p,z}\implies x-y\in\mathrm{Im}(\alpha\otimes 1_Z)$. For if $\beta(n)=\beta(n')$ then $\beta(n-n')=0$ by definition $n-n'=\alpha(m)$ for some $m\in M$. It follows that $n\otimes z-n'\otimes z=\alpha(m)\otimes z=(\alpha\otimes 1_Z)(m\otimes z)$. This shows that $\overline{\mathcal{I}_{p,z}}$ is a singleton, whose element we define to be $\psi(p,z)$. Then ψ is the unique map such that

$$\psi(\beta(n),z)=\overline{n\otimes z}$$

for all $n\in N$ and $z\in Z$. Clearly ψ is A-bilinear and therefore there is a map $\sigma: P\otimes_A Z\to(N\otimes_A Z)/\mathrm{Im}(\alpha\otimes 1_Z)$ such that $\sigma(\beta(n)\otimes z)=\overline{n\otimes z}$. By linearity $\sigma\circ\overline{\beta\otimes 1_Z}$ is the identity which shows $\overline{\beta\otimes 1_Z}$ is injective. Therefore $\ker(\beta\otimes 1_Z)=\mathrm{Im}(\alpha\otimes 1_Z)$ as required (for $x\in\ker(\beta\otimes 1_Z)\implies(\beta\otimes 1_Z)(x)=0\implies\left(\overline{\beta\otimes 1_Z}\right)(\overline{x})=0\implies\overline{x}=0\implies x\in\mathrm{Im}(\alpha\otimes 1_Z)$). ∎

Proposition 3.4.29 (Quotient of a Tensor Product)
Consider exact sequences

$$0\to M'\xrightarrow{i} M\xrightarrow{\pi_1} M/M'\to 0$$

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$$0 \to N' \xrightarrow{j} N \xrightarrow{\pi_2} N/N' \to 0$$

Then there is an exact sequence

$$\begin{array}{rcccl} (M' \otimes N) \oplus (M \otimes N') & \xrightarrow{\alpha} & M \otimes N & \xrightarrow{\pi_1 \otimes \pi_2} & M/M' \otimes N/N' \to 0 \\ (m' \otimes n, m \otimes n') & \to & i(m') \otimes n + m \otimes j(n') & \to & \overline{m} \otimes \overline{n} \to 0 \end{array}$$

In particular there is a canonical isomorphism

$$(M \otimes N)/(M' \otimes N + M \otimes N') \cong M/M' \otimes N/N'$$

Proof. Observe $\pi_1 \otimes \pi_2 = \left(\pi_1 \otimes 1_{N/N'}\right) \circ (1_M \otimes \pi_2)$ by comparison on elementary tensors. By (3.4.28) each of these maps is surjective, and so the composite is surjective. Evidently $(\pi_1 \otimes \pi_2) \circ (i \otimes 1_N) = 0$ and $(\pi_1 \otimes \pi_2) \circ (1_M \otimes j) = 0$ by checking elementary tensors. This in particular means that $(\pi_1 \otimes \pi_2) \circ \alpha = 0$ where α is the first map. Therefore $\mathrm{Im}(\alpha) \subseteq \ker(\pi_1 \otimes \pi_2)$ and it suffices to demonstrate the reverse inclusion. Suppose $z \in \ker(\pi_1 \otimes \pi_2)$. This means precisely that $(1_M \otimes \pi_2)(z) \in \ker\left(\pi_1 \otimes 1_{N/N'}\right) = \mathrm{Im}\left(i \otimes 1_{N/N'}\right)$ by (3.4.28). Therefore $(1_M \otimes \pi_2)(z) = \left(i \otimes 1_{N/N'}\right)(y)$ for $y \in M' \otimes N/N'$. By (3.4.28) again $y = (1_{M'} \otimes \pi_2)(x)$ for $x \in M' \otimes N$ and $(1_M \otimes \pi_2)(z) = (i \otimes \pi_2)(x)$. Therefore $(1_M \otimes \pi_2)(z - (i \otimes 1_N)(x)) = 0$ and $z - (i \otimes 1_N)(x) = (1_M \otimes j)(w)$ for some $w \in M \otimes N'$. Therefore we conclude $z \in \mathrm{Im}(\alpha)$ as required. ∎

3.4.6 Vector Space Tensor Product

Recall that vector spaces are free, every linearly independent set may be extended to a basis and every subspace is a direct factor. This simplifies the structure of tensor product.

Proposition 3.4.30
Let V, W be k-vector spaces with subspaces V' and W'. Then the canonical k-module homomorphism

$$V' \otimes_k W' \to V \otimes_k W$$

is injective.

Proof. The homomorphism exists by (3.4.4). By (3.3.138) both V' and W' are direct factors. Therefore the map is injective by (3.4.22). ∎

Proposition 3.4.31
Let V, W be k-modules. Suppose $\{v_i\}_{i \in I}$ and $\{w_j\}_{j \in J}$ are linearly independent (resp. bases) then $\{v_i \otimes w_j\}_{(i,j) \in I \times J}$ is a k-linearly independent subset (resp. a k-basis) of $V \otimes_k W$.

Proof. The case that they are bases is simply (3.4.24). For the general case we may use (3.3.125) to extend to bases, then a-fortiori the given set, being a subset of a basis, is linearly independent. ∎

3.4.7 Algebra Tensor Product

Let A be a commutative ring and revert to the case of commutative tensor product described in (3.4.4). We show that given two A-algebras B, C the tensor product naturally forms an algebra. We first prove some preliminary results.

Lemma 3.4.32
Let B be a commutative A-algebra. Then there exists a unique homomorphism of A-modules

$$B \otimes_A B \to B$$

such that

$$b \otimes b' \to bb'$$

Proof. Multiplication is bilinear so the map exists by universal property. ∎

Lemma 3.4.33
Let B, C be commutative A-algebras and define the A-module $X \coloneqq B \otimes_A C$. Then there is a unique homomorphism of A-modules

$$m : X \otimes_A X \to X$$

such that

$$(b \otimes c) \otimes (b' \otimes c') \to (bb') \otimes (cc')$$

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Define $1_X := (1 \otimes 1)$ *then it satisfies*

a) $m(1_X \otimes x) = m(x \otimes 1_X) = x$
b) $m(x \otimes y) = m(y \otimes x)$
c) $m(x \otimes m(y \otimes z)) = m(m(x \otimes y) \otimes z)$
d) $m((x + y) \otimes z) = m(x \otimes z) + m(y \otimes z)$
e) $m(x \otimes (y + z)) = m(x \otimes y) + m(x \otimes z)$

for $x, y, z \in X$.

Proof. By associativity and commutativity of the tensor product there is an A-module isomorphism

$$X \otimes_A X \cong (B \otimes_A B) \otimes_A (C \otimes_A C)$$
$$(b \otimes c) \otimes (b' \otimes c') \to (b \otimes b') \otimes (c \otimes c')$$

composing with $m_B \otimes m_C$ where $m_B : (B \otimes_A B) \to B$ and $m_C : (C \otimes_A C) \to C$ are the maps given in (3.4.32), yields the required map m.

The properties may be verified on tensors of the form $x_i = (b_i \otimes c_i)$ and $y_j = \left(b'_j \otimes c'_j\right)$. The results follow from linearity, since $\left(\sum_i x_i\right) \otimes \left(\sum_j y_j\right) = \sum_{i,j}(x_i \otimes y_j)$. ∎

Proposition 3.4.34 (Algebra Tensor Product)
Let B, C *be commutative* A *algebras. Then the* A*-module* $B \otimes_A C$ *has a unique commutative ring structure with unit* $1 \otimes 1$ *and multiplication which satisfies*

$$\cdot : (B \otimes_A C) \times (B \otimes_A C) \to B \otimes_A C$$
$$(b \otimes c) \cdot (b' \otimes c') := (bb') \otimes (cc')$$

and may be extended by linearity. Further there is an ring homomorphism

$$i_A : A \to B \otimes_A C$$
$$a \to (a \otimes 1) = (1 \otimes a)$$

making $B \otimes_A C$ *into an* A*-algebra.*

Proof. Let $X := B \otimes_A C$ and consider the map $m : X \otimes_A X \to X$ defined in (3.4.33). Define $x \cdot y := m(x \otimes y)$ then the properties of m ensure that this satisfies the properties of a ring. ∎

Proposition 3.4.35 (Criteria to be an algebra homomorphism)
Let B, C, Z *be commutative* A*-algebras and let* $\phi : B \otimes_A C \to Z$ *be an* A*-module homomorphism. Then the following are equivalent*

a) ϕ *is an* A*-algebra homomorphism*
b) $\phi((b \otimes c) \cdot (b' \otimes c')) = \phi(b \otimes c) \cdot \phi(b' \otimes c')$

Similarly suppose $\psi : Z \to B \otimes_A C$ *is an* A*-module homomorphism and* Z *is generated as an* A*-module by* $S \subset Z$*. Then the following equivalent*

a) ψ *is an* A*-algebra homomorphism*
b) $\psi(ss') = \psi(s)\psi(s')$

Proof. In each case $a) \implies b)$ is obvious. For the converse we simply need to show that ϕ and ψ are in general multiplicative. This follows by linearity and because every element of the tensor product is a linear combination of elementary tensors. ∎

Proposition 3.4.36 (Functoriality)
Let $\phi : B \to B'$ *and* $\psi : C \to C'$ *be* A*-algebra homomorphisms. The* A*-module homomorphism*

$$\phi \otimes \psi : B \otimes_A C \to B' \otimes_A C'$$
$$b \otimes c \to \phi(b) \otimes \phi(c)$$

is an A*-algebra homomorphism.*

Proof. The A-module homomorphism exists by (3.4.4). It is in A-algebra homomorphism by (3.4.35). ∎

Proposition 3.4.37 (Tensor Product is Coproduct)
Let B, C *be commutative* A*-algebras then there are* A*-algebra homomorphisms*

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$$\begin{aligned} i_B : B &\to B \otimes_A C \\ b &\to b \otimes 1 \\ i_C : C &\to B \otimes_A C \\ c &\to 1 \otimes c \end{aligned}$$

which are natural in B and C respectively. In particular $B \otimes_A C$ is both a C-algebra and a B-algebra.

Furthermore for Z an A-algebra there is a bijection

$$\begin{aligned} \mathrm{AlgHom}_A(B \otimes_A C, Z) &\cong \mathrm{AlgHom}_A(B, Z) \times \mathrm{AlgHom}_A(C, Z) \\ \psi &\to (\psi \circ i_B, \psi \circ i_C) \end{aligned}$$

which is natural in Z.

Proof. The existence of i_B and i_C is easily demonstrated using universal property of tensor product. Observe that in general

$$(\psi \circ i_B)(b) = \psi(b \otimes 1_C)$$

$$(\psi \circ i_C)(c) = \psi(1_B \otimes c)$$

Furthermore the map is clearly well-defined. It's injective because if ψ and ψ' have the same image then $\psi(b \otimes 1) = \psi'(b \otimes 1)$ and $\psi(1 \otimes c) = \psi'(1 \otimes c)$ whence $\psi(b \otimes c) = \psi'(b \otimes c)$. By linearity they are everywhere equal.

To show that the map is a bijection we construct a two-sided inverse. For given A-algebra homomorphisms $f : B \to Z$ and $g : C \to Z$ then $(b, c) \to (f(b)g(c))$ is A-bilinear and so corresponds to an A-module homomorphism

$$\begin{aligned} \psi_{f,g} : B \otimes_A C &\to Z \\ b \otimes c &\to f(b)g(c) \end{aligned}$$

and by (3.4.35) this is an algebra homomorphism. Clearly $\psi \circ i_B = f$ and $\psi \circ i_C = g$. Conversely let $\psi : B \otimes_A C \to Z$ be an algebra homomorphism and we define $f(b) \coloneqq \psi(b \otimes 1_C)$ and $g(c) \coloneqq \psi(1_C \otimes c)$. Then $f(b)g(c) = \psi(b \otimes c)$ which shows that $\psi_{f,g}$ agrees with ψ on elementary tensors, and therefore is identically equal. This completes the demonstration that $\psi_{f,g}$ is a two-sided inverse to the given map.

Naturality in Z is straightforward. ∎

Proposition 3.4.38 (Extension of Scalars)
Let B, C be commutative A-algebras. Then define the C-algebra

$$B_{(C)} \coloneqq C \otimes_A B$$

There is a natural isomorphism for any C-algebra Z

$$\begin{aligned} \mathrm{AlgHom}_C\left(B_{(C)}, Z\right) &\cong \mathrm{AlgHom}_A(B, Z) \\ \psi &\to \psi \circ i_B \end{aligned}$$

Proof. We consider the commutative diagram obtained from (3.4.37)

$$\begin{array}{ccc} \mathrm{AlgHom}_A\left(B_{(C)}, Z\right) & \xrightarrow{\sim} & \mathrm{AlgHom}_A(B, Z) \times \mathrm{AlgHom}_A(C, Z) \\ \cup & & \cup \\ \mathrm{AlgHom}_C\left(B_{(C)}, Z\right) & \xrightarrow{\sim} & \mathrm{AlgHom}_A(B, Z) \times \{i_{CZ}\} \end{array}$$

An A-algebra homomorphism $\phi : B_{(C)} \to Z$ is a C-algebra homomorphism precisely when $\phi \circ i_C = i_{CZ}$ so that the bottom arrow is well-defined and bijective as required. ∎

Proposition 3.4.39 (Transitivity of Extension of Scalars)
Let B, C be commutative A-algebras and D a commutative C-algebra. Then there is an isomorphism of D-algebras

$$\begin{aligned} D \otimes_C (C \otimes_A B) &\cong D \otimes_A B \\ d \otimes (c \otimes b) &\to (dc) \otimes b \end{aligned}$$

Proof. Let Z be a D-algebra then there is a natural isomorphism of functors

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$$\begin{aligned}\mathrm{AlgHom}_D(D\otimes_C(C\otimes_A B),Z) &\cong \mathrm{AlgHom}_C(C\otimes_A B,Z)\\ &\cong \mathrm{AlgHom}_A(B,Z)\\ &\cong \mathrm{AlgHom}_D(D\otimes_A B,Z)\end{aligned}$$

Consider the case $Z = D\otimes_A B$ and the identity map yields the required isomorphism (by the Yoneda Lemma). ∎

Proposition 3.4.40
Let B be a commutative A-algebra then there is a natural isomorphism of A-algebras

$$\begin{aligned}u_B : B &\xrightarrow{\sim} B_{(A)}\\ b &\to 1\otimes b\\ ab &\leftarrow a\otimes b\end{aligned}$$

Furthermore for C a commutative A-algebra there is a commutative diagram

$$\begin{array}{ccc} B & \xrightarrow{u_B} & A\otimes_A B \\ & \searrow^{u_B} & \downarrow{\scriptstyle i_A\otimes 1_B} \\ & & C\otimes_A B \end{array}$$

Proof. By (3.4.38) there is an bijection natural in Z

$$\begin{aligned}\mathrm{AlgHom}_A\left(B_{(A)},Z\right) &\xrightarrow{\sim} \mathrm{AlgHom}_A(B,Z)\\ \psi &\to \psi(1\otimes -)\end{aligned}$$

which by the Yoneda Lemma (2.6.58) yields the required isomorphism (given by the image of $1_{B_{(A)}}$). ∎

Proposition 3.4.41 (Structural Morphisms are Injective)
Let B, C be commutative A-algebras such that the structural morphisms are injective and A is a direct factor of B and C (e.g. if $A = k$ is a field). Then the canonical maps (3.4.37)

$$B \to B\otimes_A C$$

$$C \to B\otimes_A C$$

are injective.

Proof. By (3.4.22) the canonical map

$$B\otimes_A A \to B\otimes_A C$$

is injective. By (3.4.40) we have a canonical isomorphism $B\cong B\otimes_A A$ and the composite is simply the canonical map $B\to B\otimes_A C$. ∎

Proposition 3.4.42 (Extension of Ideals under Tensor Product)
Let B, C be commutative A-algebras and $\mathfrak{b}\lhd B$ and $\mathfrak{c}\lhd C$ ideals. Then $\mathfrak{b}^e$ (resp. $\mathfrak{c}^e$) is the image of the A-module $\mathfrak{b}\otimes_A C$ (resp. $B\otimes_A \mathfrak{c}$) in $B\otimes_A C$. Furthermore there is a canonical A-algebra isomorphism

$$(B\otimes_A C)/(\mathfrak{b}^e+\mathfrak{c}^e)\cong B/\mathfrak{b}\otimes_A C/\mathfrak{c}$$

Proof. Let $\phi : B\to B\otimes_A C$ be the structural morphism then for $b\in\mathfrak{b}$ we have $b\otimes c = (c\otimes 1)(b\otimes 1) = (1\otimes c)\phi(b)$ whence $\phi(\mathfrak{b})\subseteq \mathfrak{b}\otimes_A C\subseteq \mathfrak{b}^e$. Evidently $\mathfrak{b}\otimes_A C$ is an ideal and therefore we see $\mathfrak{b}\otimes_A C = \mathfrak{b}^e$ as required. The isomorphism follows from (3.4.29). ∎

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